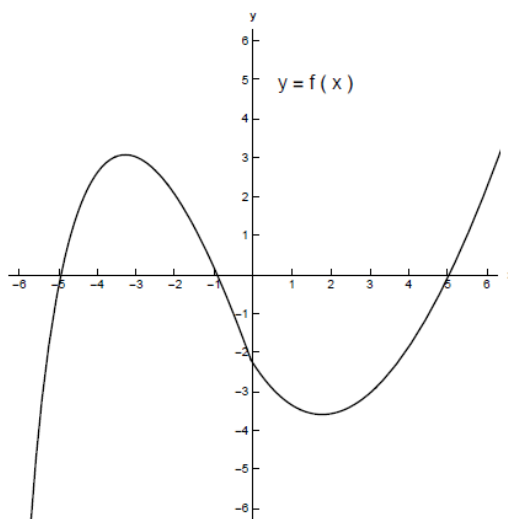


Problem 5

Problem 5

Problem 1 Suppose we are given the graph of a function f :



(a) Use this graph to find the following: (Assume all values will be integers or $+\infty$ or $-\infty$)

(i) all x where $f(x) = 0$,

Explanation. $f(x)$ is zero when the function crosses the x -axis. Therefore $f(x) = 0$ when $x = -5$, $x = -1$, and $x = 5$.

(ii) all x where $f(x) > 0$,

Explanation. $f(x)$ is positive when the graph of the function is above the x -axis. Therefore $f(x) > 0$ on $(-5, -1) \cup (5, \infty)$.

(iii) all x where $f(x) < 0$, and

Explanation. $f(x)$ is negative when the graph of the function is below the x -axis. Therefore $f(x) < 0$ on $(-\infty, -5) \cup (-1, 5)$.

(iv) all x where $f(x)$ attains a local maximum and all x where f attains a local minimum.

Explanation. $f(x)$ has a local maximum at $x = -3$. $f(x)$ has a local minimum at $x = 2$.

Without sketching the graph of f' find

- (b) (i) all x where $f'(x) = 0$,

Explanation. $f'(x)$ is zero when the tangent line has a slope of zero, which is approximately at $x = -3$ and $x = 2$. Note, for this question, these are the same answers as the (local) highest and lowest point for the graph of f .

- (ii) all x where $f'(x) > 0$,

Explanation. $f'(x)$ is positive when the slope of the tangent line is positive. Observe that f is increasing on $(-\infty, -3)$, $(2, \infty)$ and this same set of intervals is where the tangent lines have positive slope. Therefore $f'(x) > 0$ on $(-\infty, -3)$, $(2, \infty)$.

- (iii) all x where $f'(x) < 0$, and

Explanation. $f'(x)$ is negative when the slope of the tangent line is negative. Observe that f is decreasing on $(-3, 2)$ and this interval is where the tangent lines have negative slope. Therefore $f'(x) < 0$ on $(-3, 2)$.

- (iv) On the following intervals, does $f'(x)$ seem to be increasing or decreasing?

- i. $(-\infty, 0)$

Explanation. decreasing

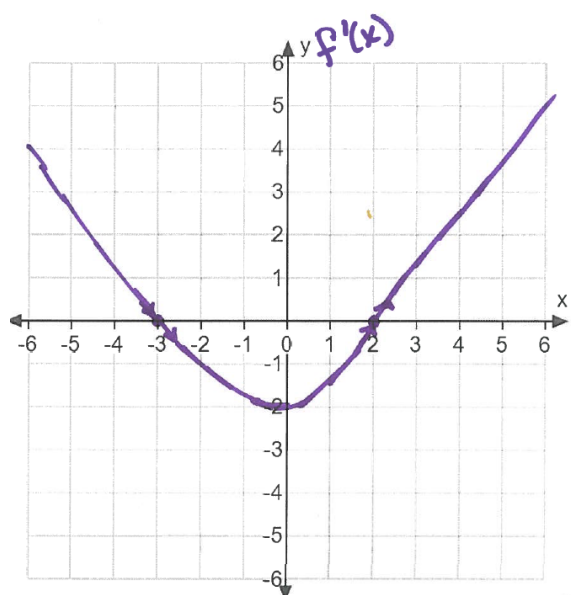
- ii. $(0, \infty)$

Explanation. increasing

- (c) Sketch a graph of f' .

Explanation. The graph of f' is approximately

Problem 5



[2]